A **Mini** Project report On

Customer reorder prediction Using Machine Learning

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Chapter 1: Introduction

1 Introduction

Customer reorder prediction is an important problem in retail and e-commerce because it helps businesses understand customer buying behavior and improve product recommendations, inventory planning, and personalized marketing. In online grocery platforms like Instacart, predicting whether a product will be reordered can support better demand forecasting and customer retention strategies.

The Instacart Market Basket Analysis dataset contains customer order histories, product details, and reorder information. Since the target variable reordered has two possible values, 0 and 1, this problem is a binary classification problem. Initially, simpler approaches and limited files gave misleading results, especially when regression-based methods appeared to perform well. However, after restructuring the project using prior order history and proper classification models, the task became more realistic and academically correct.

This project aims to compare multiple machine learning algorithms and identify the model that best predicts product reorders. Special attention was given to preprocessing, feature engineering, handling class balance, and evaluating models using not only accuracy but also precision, recall, F1-score, confusion matrix, ROCAUC, PR-AUC, and cross-validation

1.2Importance of the Topic

This project focuses on predicting whether a customer will reorder a product using the Instacart Market Basket Analysis dataset. The task was treated as a binary classification problem where the target variable indicates whether a product is reordered or not. Multiple machine learning algorithms were applied, including Logistic Regression, Decision Tree, Naive Bayes, SVM, KNN, Random Forest, Gradient Boosting, HistGradientBoosting, XGBoost, and KMeans for comparison. Data preprocessing, feature engineering, train-test splitting, balanced sampling, and performance evaluation were carried out. Among all the tested models, HistGradientBoosting achieved the best accuracy of 68.84%, while XGBoost showed stable cross-validation performance. The results show that boosting-based ensemble methods are the most effective for this reorder prediction task.

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Chapter 2: Proposed Methodology

The project was completed in the following steps:

Step 1 : Data Collection and Loading

The dataset was downloaded using kagglehub from the instakart market basket analysis dataset. The project was carried out using the Instacart dataset. The main files used were:

- orders.csv
- order_products__prior.csv
- order_products__train.csv

These files were selected because reorder prediction depends heavily on previous purchase history. The prior orders were used to build customer and product behavior features, while the train orders were used as the final labeled dataset for prediction.

The first step was data loading and merging. The orders were separated into prior and train sets using the eval_set column. Then prior_data and train_data were created by merging product-order information with order metadata

Step 2 : Data Understanding

Because the full dataset was large and caused memory issues in Colab, a reduced but manageable version of the data was used:

- prior_data: 400,000 rows
- train_data: 150,000 rows
- final balanced modeling sample: 60,000 rows

After this, preprocessing was performed. Missing values were filled using suitable methods such as median filling and replacing missing days since prior order with 0. Identifier columns like order_id, user_id, and eval_set were removed before modeling because they do not directly help prediction.

Step 3 : Exploratory Data Analysis

Exploratory Data Analysis was performed to understand class distribution, order hour trends, cart position distribution, correlation between numeric features, and possible outliers. Outliers were checked using boxplots and IQR-based counts, but they were not removed blindly because many extreme values reflect real customer behavior.

Step 4 : Data Preprocessing

After this, preprocessing was performed. Missing values were filled using suitable methods such as median filling and replacing missing days since prior order with 0. Identifier columns like order_id, user_id, and eval_set were removed before modeling because they do not directly help prediction

Step 5 : Model

For model training, a balanced sampled dataset of 60,000 rows was created, with 30,000 reordered rows and 30,000 non-reordered rows. Then the data was split into training and testing sets:

- training set: 48,000 rows
- test set: 12,000 rows

train_test_split with stratify=y was used to maintain class balance. Feature scaling with StandardScaler was applied for models that require scaled data, such as Logistic Regression, SVM, KNN, and Naive Bayes.

The following models were applied:

- Dummy Baseline
- Logistic Regression
- Class-Weighted Logistic Regression
- Decision Tree
- Naive Bayes
- SVM
- KNN
- Random Forest
- Gradient Boosting
- HistGradientBoosting
- XGBoost
- KMeans for experimental comparison

Step 6 : Hyperparameter Tuning

Hyperparameter tuning was done manually for ensemble models by adjusting parameters such as n_estimators, max_depth, learning_rate, subsample, and min_samples_leaf. This was light manual tuning, not full GridSearchCV or RandomizedSearchCV

Bagging and boosting were also explored. Random Forest represented bagging, while Gradient Boosting, HistGradientBoosting, and XGBoost represented boosting. In this project, boosting performed better than bagging, showing that sequential error correction was more effective than simple parallel tree aggregation for reorder prediction.

Results/Outputs

Model	Accuracy
HistGradientBoosting	0.688417
XGBoost	0.687083
Gradient Boosting	0.686667
Random Forest	0.679167
Class-Weighted Logistic Regression	0.665167
Logistic Regression	0.665167

SVM	0.664917
KNN	0.653583
Decision Tree	0.641500
KMeans	0.583750
Naive Bayes	0.525167
Dummy Baseline	0.500000

The best-performing model was HistGradientBoosting, with an accuracy of 68.84 percent. XGBoost and Gradient Boosting gave very similar results and were also strong performers. Random Forest performed reasonably well, but boosting models were better overall. Simpler models such as Logistic Regression and SVM performed moderately, while Naive Bayes gave the weakest result among the classification models.

For the best model, additional evaluation metrics were also computed:

- **HistGradientBoosting ROC-AUC: 0.7529**
- **HistGradientBoosting PR-AUC: 0.7568**

These values show that the model was better than what accuracy alone suggests and had decent discriminative ability.

XGBoost cross-validation was also performed:

- **CV scores: [0.6814, 0.6768, 0.6842, 0.6803, 0.6778]**
- **Mean CV accuracy: 0.6801**
- **Standard deviation: 0.0026**

This indicates that the model's performance was consistent across folds and not due to a lucky split.

Feature importance from XGBoost showed the top five features as:

- order_number
- user_product_reorder_rate
- product_reorder_rate
- add_to_cart_order
- user_reorder_ratio

This means that reorder prediction depends strongly on order history, the relationship between a user and a product, and overall reorder trends of products.

A very important observation from the project is that Linear Regression is not suitable for this task, even if it appears to give a good score after thresholding. Since reordered is a binary target, classification algorithms are the correct choice. Similarly, KMeans was included only as an experimental comparison and should not be treated as a proper classifier.

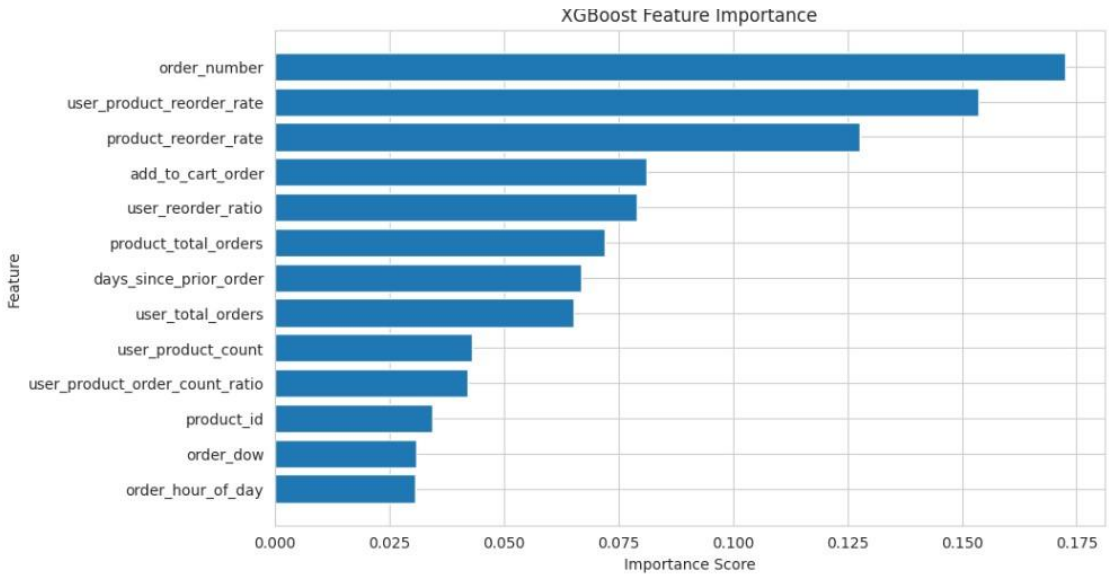
Why the Accuracies Are Moderate

ct are moderate rather than extremely high because customer reorder prediction is a difficult realstructured ning problem. Customer purchasing behavior is not fully determined by only a few variables er, cart position, and reorder rate. In practice, product reordering depends on al preference changes, seasonal demand, promotions, stock availability, brand switching, and one-time purchases. S factors are not fully present in the dataset, the models cannot capture these customer behavior perfectly.

curacy is that the project was executed on a reduced sample of the original dataset due to memory limitations ir Colab. Although balanced sampling was used to maintain fairness between reordered and nondatased classes, the model still had access to less historical information than the full - provide. More compl order history would likely improve the learned patterns.

The evaluation setup was also intentionally kept academically correct. Features were built mainly from prior orders and the prediction was made on train data without relying on information leakage. This makes the results more realistic, but it also prevents artificially inflated accuracy. In many machine learning projects, higher accuracy may appear when the pipeline is less strict, but such results are less reliable.

It is also important to note that accuracy alone does not tell the full story. In this project, the best model, HistGradientBoosting, achieved not only the highest accuracy but also a ROC-AUC of 0.7529 and a PR-AUC of 0.7568, which shows that the model still has good predictive ability. Therefore, the moderate accuracy should not be seen as failure, but as a realistic result for a complex behavioral prediction problem using limited observable features.



```
XGBoost Accuracy: 0.6870833333333334

Classification Report:
              precision    recall  f1-score   support

           0       0.68      0.71      0.69      6000
           1       0.70      0.66      0.68      6000

    accuracy                           0.69     12000
   macro avg       0.69      0.69      0.69     12000
weighted avg       0.69      0.69      0.69     12000


Confusion Matrix:
 [[4276 1724]
 [2031 3969]]
```

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HistGradientBoosting ROC-AUC: 0.75292425
HistGradientBoosting PR-AUC: 0.7567679433243116
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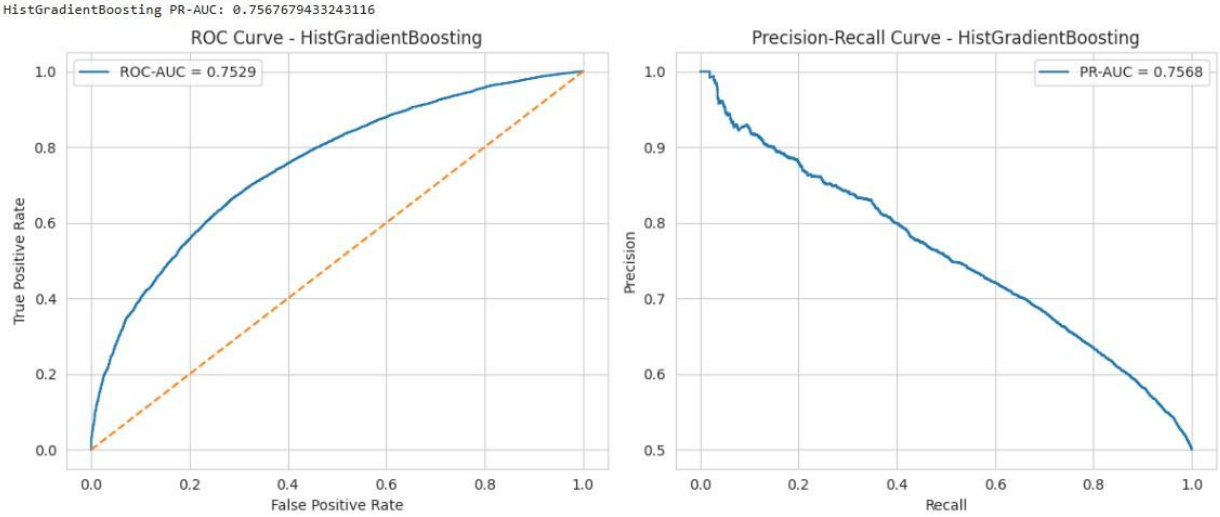

Conclusion

This project implemented a machine learning solution for customer product reorder prediction using the Instacart dataset. The problem was correctly treated as a binary classification task, and multiple models were trained and compared. After preprocessing, feature engineering, balanced sampling, and evaluation, HistGradientBoosting was found to be the best-performing model with an accuracy of 68.84 percent. XGBoost and Gradient Boosting also performed strongly, showing that boosting-based ensemble methods are the most suitable for this problem. The results also showed that user-product

interaction features and reorder history are highly important for prediction. Overall, the project demonstrates that machine learning can effectively model customer reorder behavior, even though accuracy can still be improved with more data and richer features.

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Future Score

In future work, the project can be improved by using the full prior dataset, creating stronger recencybased features, and applying more systematic hyperparameter tuning such as GridSearchCV. Additional advanced models and more computational resources may further improve performance.

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Chapter 4: References

- Instacart Market Basket Analysis Dataset, Kaggle
- Scikit-learn Documentation
- XGBoost Documentation
- Research articles and tutorials on ensemble learning, bagging, and boosting